

Joseph Damilare Chinedum Ukeje

Senior Software Engineer | Java/J2EE | Spring Boot | Payment Platforms

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Summary

Senior Software Engineer with 6+ years of continuous experience delivering enterprise Java applications and cloud-based payment platforms. Deep expertise in **Java 21/17/8, Java/J2EE, Spring Boot 3.x, design patterns, custom frameworks, distributed systems, and high-throughput transaction processing**. Owns complex product areas end to end, from requirements and architecture through testing, optimisation, release, and production support. Delivered ISO 20022/SWIFT infrastructure, Open Banking and AML/KYC integrations, secure REST/gRPC APIs, Kafka workflows, and observable Kubernetes services at Tier-1 bank transaction volumes. Provides peer leadership through code review, technical training, automation, process improvement, and pragmatic architectural decisions grounded in collaboration, ownership, results orientation, integrity, and innovation.

Technical Skills

- **Languages:** Java (21, 17, 8), Java/J2EE, SQL, Python, TypeScript, JavaScript, Solidity
- **Java Frameworks:** Spring Boot 3.x, Spring WebFlux, Spring Data JPA/Hibernate, Spring Security, Spring AOP/AspectJ, Spring Batch, Spring AMQP, Spring Authorization Server, Jakarta EE / Java EE, MapStruct, Lombok, Quartz Scheduler
- **Code Sophistication & Abstractions:** Object-oriented design, SOLID, GoF design patterns, Clean Architecture, strategic/tactical DDD, CQRS, custom framework design, reusable abstractions, third-party framework adaptation
- **Performance & Memory:** Java 21 Virtual Threads, CompletableFuture, ExecutorService, multithreading, JVM heap/GC tuning, async-profiler, HikariCP, query optimisation, Redis/Caffeine caching, P99 latency analysis
- **APIs & Enterprise Integration:** REST APIs, gRPC-Java, Protobuf, GraphQL, SOAP, Open Banking APIs, Netty TCP, IBM MQ, JMS, API versioning, contract-first design
- **Distributed Systems:** Microservices, event-driven architecture, eventual consistency, idempotency, atomic transactions, high availability, fault tolerance, graceful degradation, resilience patterns
- **Messaging:** Kafka partition strategy, consumer groups, exactly-once semantics, Schema Registry and DLQ; RabbitMQ, Azure Service Bus, AWS SNS/SQS, Spring AMQP
- **Databases & Caching:** PostgreSQL, Oracle, SQL Server, MySQL, MongoDB, Redis Cluster, data modelling, indexing, query optimisation, Pub/Sub, distributed locks, TTL eviction
- **Testing & Quality:** Manual, automated, unit, integration, end-to-end, contract and performance testing; TDD/BDD, JUnit 5, Mockito, WireMock, Testcontainers, Postman
- **Code Review & Standards:** In-depth review of structure, security, performance, reliability, maintainability and testability; architecture decision records and quality gates
- **Developer Tooling & Automation:** IntelliJ IDEA, Maven, Gradle, Git, GitHub, CI/CD, reusable development utilities, workflow automation and release automation, Jira, Confluence
- **Cloud & DevOps:** AWS (EC2, EKS, S3, Lambda, SNS, SQS, CloudWatch), Azure (AKS, App Service, Functions, Service Bus, DevOps, Key Vault), GCP, Docker, Kubernetes, Helm, Terraform, Jenkins, GitHub Actions
- **Security & Observability:** OAuth 2.0, JWT, RBAC, OWASP, Vault, Veracode, SonarQube, Prometheus, Grafana, Splunk, CloudWatch, Micrometer, structured logging, metrics, tracing and alerting
- **Payments & Financial Services:** ISO 20022/SWIFT, pacs.008/pacs.002/pacs.028, digital payments, Open Banking, payment onboarding, AML/KYC, auditability and reconciliation
- **Technical Leadership:** End-to-end ownership, peer leadership, mentoring, technical training, process improvement, architectural discussions, stakeholder communication and Agile/Scrum
- **Product & UI Collaboration:** React.js, TypeScript, Redux Toolkit, usability, secure UI integration, responsive design, aesthetic standards, Product/QA/UX collaboration
- **Values:** Collaboration, ownership, results orientation, integrity, innovation, initiative, pragmatism, policy adherence, clear written and verbal communication

Professional Experience

Senior Software Engineer (Backend)

Interswitch Group

Apr 2025 - Present

Remote

- Engineered end-to-end **ISO 20022/SWIFT payment flows** for pacs.008, pacs.002, and pacs.028 with XML canonicalisation, digital signatures, and namespace standardisation - eliminating **100% of production payment failures** at Tier-1 bank transaction volumes.
- Owned the architecture, implementation, and production lifecycle of a centralised **audit service for Interswitch's core banking infrastructure**, capturing immutable transaction and user-action trails across distributed services for compliance, investigations, and end-to-end traceability.
- Integrated **Open Banking and AML/KYC provider APIs** into payment onboarding and account workflows, implementing request orchestration, retry and timeout handling, and response validation - reducing manual compliance review time by roughly **40%**.
- Developed **IRIS, Interswitch's proprietary core banking payment gateway and inter-bank transfer engine**, on a custom Netty TCP server; managed connection lifecycle, protocol framing, session state, idempotency, and atomic transaction boundaries across high-concurrency payment flows.

- Owned broad areas of **Java/J2EE and Spring-based REST APIs** for **10+ institutions** across **30+ production endpoints**, translating business requirements into maintainable multi-tenant core-banking services.
- Architected **gRPC-Java service integrations** for blockchain RPC validation, authoring Protobuf contracts and engineering serialisation, streaming, and graceful degradation across distributed Java microservices.
- Performed **root-cause analysis on failed and stalled transactions** using service logs and end-to-end transaction trails, isolating faults across service and provider-integration boundaries and restoring reliable payment processing.
- Built **Python ETL and automation scripts** to reconcile transaction and settlement data across internal core banking systems, reducing manual reconciliation effort and surfacing discrepancies for operations teams.
- Orchestrated IRIS connection state with **CompletableFuture-based Java 8 multithreading and Java 21 Virtual Threads** - sustaining **5,000+ concurrent connections per node and 10,000+ concurrent threads** while processing millions of settlement transactions at peak load.
- Conceived and implemented **PRM, Interswitch's proprietary payment-risk framework**, as a non-intrusive Spring AOP/AspectJ layer with reusable custom annotations, CQRS patterns, and graceful degradation - reducing fraud-related production incidents by roughly **35%**.
- Designed and shipped **Spring Boot microservices on Kubernetes** against sub-100ms latency targets; integrated Kafka and Azure Service Bus pipelines across **5+ distributed Java services** to improve asynchronous throughput and reliability.
- Engineered **automated stress-testing tooling using production-traffic replay** to expose performance bottlenecks, validate failure behaviour, and establish system-capacity thresholds before release.
- Designed **Redis caching and concurrency patterns** using cluster mode, distributed locks for payment deduplication, and TTL eviction aligned with financial-data residency requirements - reducing database round-trips and cutting P99 latency by roughly **30%**.
- Designed **shared infrastructure components**, including reusable caching libraries and circuit-breaker systems, to standardise fault isolation, graceful degradation, and recovery during downstream failures.
- Implemented OAuth 2.0 and JWT with **Spring Authorization Server**, built Prometheus/Grafana observability around DORA metrics, and resolved **100% of Veracode high-severity vulnerabilities**.
- Built secure **React.js and TypeScript internal operations dashboards** for Open Banking and payment workflows, using Redux Toolkit, reusable components, role-aware views, searchable transaction trails, and protected API integrations.
- Contributed to Interswitch's internal **AI Gateway** on LiteLLM, providing authentication, rate limiting, and LLM routing for **50+ engineers**; integrated Claude Code while retaining ownership of design, verification, security, and code quality.
- Built **Vera AI**, an agentic onboarding assistant using **Spring AI's ChatClient** abstraction within the Java/Spring Boot stack - reducing new-hire onboarding time by roughly **25%**.
- Performed in-depth code reviews covering structure, performance, memory use, reliability, security, maintainability, and testability; led platform-wide architecture discussions and documented decisions through ADRs.
- Provided peer leadership and technical training to junior engineers, independently identifying platform problems, proposing pragmatic solutions, and following changes through implementation and production verification.
- Partnered with Product Owners, architects, QA, operations, and compliance stakeholders to balance customer value, security, performance, delivery speed, and long-term platform maintainability.
- Tech Stack:** Java 21, Spring Boot 3.x, Spring WebFlux, Spring Security, Spring AOP/AspectJ, Spring AI, Netty TCP, gRPC-Java, Kafka, Azure Service Bus, Redis, PostgreSQL, Python, React.js, TypeScript, Docker, Kubernetes

Senior Software Engineer
Huawei Technologies Nigeria

Jun 2024 - Apr 2025
Nigeria

- Designed scalable **Java/Spring Boot microservices** for Huawei Cloud's enterprise B2B platform, architecting REST controllers, service layers, and asynchronous resource-provisioning workflows.
- Built Spring Boot services with **Spring Data JPA/Hibernate** for ORM-based multi-tenant data management and implemented **Kafka ETL pipelines** for reliable, high-throughput data flow across platform services.
- Delivered a partner rewards administration platform end-to-end across **Spring Boot APIs and React/TypeScript/Redux workflows** and architected Azure DevOps CI/CD pipelines that reduced deployment time by **25%**.
- Implemented **OAuth 2.0 and JWT** authentication and authorization across Huawei Cloud's B2B microservices, securing service-to-service and partner-facing API access.
- Architected an asynchronous **RabbitMQ notification pipeline** across 5+ services with broker replicas, automatic failover, and a dead-letter queue for failed-delivery retry - achieving **99% uptime**.
- Implemented Helm charts for Kubernetes deployments, supporting efficient resource allocation, horizontal pod autoscaling, and consistent infrastructure-as-code standards on AWS EKS.
- Established manual, automated, unit, integration, and contract-testing standards with **JUnit 5, Mockito, WireMock, and Testcontainers**, achieving a **95% bug-free release rate** and accelerating delivery by **20%**.
- Mentored and technically trained junior engineers on Spring Boot microservice patterns, clean architecture, performance, testing, and code review, accelerating onboarding and raising squad-wide quality.
- Improved team delivery practices through contract-first API design, reusable CI/CD templates, peer reviews, and clearer engineering standards across application and platform work.
- Tech Stack:** Java, Spring Boot, Spring Data JPA, Hibernate, OAuth 2.0, JWT, Kafka, RabbitMQ, PostgreSQL, React.js, TypeScript, Redux, AWS EKS, Kubernetes, Helm, Azure DevOps

Software Engineer
Vision Forge AI Automations

Jan 2024 - Jun 2024
Remote, Nigeria

- Independently architected a **Spring Boot microservice integration layer** connecting 5 distributed services through Spring AMQP and RabbitMQ, reducing QA data errors by **50%** through contract validation and idempotency patterns.
- Implemented RabbitMQ event-driven messaging for asynchronous inter-service communication and refactored Spring Boot services around maintainability and clean architecture, reducing average response times by **15%**.
- Built **Java/Spring Boot APIs** for internal automation workflows, using CompletableFuture and ExecutorService for asynchronous processing and reliable integrations consumed by operational tools.
- Expanded **JUnit 5 and Mockito** coverage to **85%+** with Testcontainers and WireMock, significantly reducing regression defects across Spring Boot services.
- Architected a **GraphQL (Apollo) API layer in Java** consumed by React/TypeScript dashboards, enabling flexible queries across heterogeneous microservice data sources with DataLoader batching.
- **Tech Stack:** Java, Spring Boot, Spring AMQP, RabbitMQ, GraphQL, JUnit 5, Mockito, Testcontainers, WireMock

Software Engineer

Schlumberger (SLB) Oil & Gas Servicing Limited

Jan 2023 - Dec 2023

Lagos, Nigeria

- Designed and delivered production-grade **Java/Spring Boot REST APIs** for enterprise applications, implementing JWT, OAuth 2.0, and Twilio 2FA in line with OWASP Top 10 standards.
- Architected GCP infrastructure and Java CI/CD deployment pipelines, reducing cloud infrastructure costs by **18%** through right-sizing and environment optimisation.
- Implemented **Spring Security RBAC** across internal service boundaries and designed PostgreSQL relational models supporting high-volume operational reporting workflows.
- Built React/TypeScript operational interfaces using reusable components, Redux state management, resilient error handling, CSRF protection, and XSS-safe rendering; contributed feedback that improved usability and consistent aesthetic standards.
- Led code reviews and architecture discussions to standardise graceful degradation, testability, and architecture decision records across squads, reducing production incidents and improving cross-team transparency.
- **Tech Stack:** Java, Spring Boot, Spring Security, PostgreSQL, GCP, React.js, TypeScript, Redux

Software Engineer

The Intrepid Technologies (Chevron)

Jan 2022 - Dec 2022

Lekki, Nigeria

- Re-engineered **Java/Spring Boot backend services** for 6 enterprise applications, redesigning API response pipelines and reducing average response times by **200ms**.
- Implemented HikariCP connection pooling and query optimisation against Oracle and PostgreSQL, improving database throughput by **35%**.
- Engineered RPA-integrated Java analytics modules that eliminated manual data collation across delivery teams and reduced reporting cycle time by **40%**.
- Enhanced **Spring Batch** jobs with execution-time tracking, structured logging, and operational metrics; developed an automated purge framework removing more than **500,000 expired records daily**, improving storage utilisation and reducing operational overhead.
- Strengthened production observability with **Prometheus, Grafana, Splunk, and AWS CloudWatch**; investigated high-severity incidents through RCA, met SLA commitments, and implemented lasting corrective and preventive actions.
- Increased automated test coverage from **20% to 85%** through TDD and BDD practices using JUnit, Mockito, and Testcontainers, improving release confidence and regression protection.
- Contributed to LLD discussions, technical design reviews, sprint planning, and architectural decisions in an Agile Scrum environment; partnered with Product Owners, QA, DevOps, and architects while reviewing code and mentoring junior engineers on Java, debugging, and testing practices.
- **Tech Stack:** Java, Spring Boot, Spring Batch, HikariCP, Oracle, PostgreSQL, Prometheus, Grafana, Splunk, AWS CloudWatch, JUnit, Mockito, Testcontainers

Software Engineer (Java Developer)

Upwork

Jan 2020 - Dec 2021

Remote, Nigeria

- Designed, developed, and maintained scalable **Java/Spring Boot microservices** for international clients using Spring WebFlux, REST APIs, and Kafka for asynchronous, event-driven processing.
- Integrated **AWS Lambda, Amazon SNS, and Amazon SQS** with Spring Boot services, enabling serverless processing and reliable asynchronous communication across distributed client applications.
- Delivered JPA data models, API layers, and cloud-hosted deployment pipelines end-to-end while collaborating with client product, QA, and operations stakeholders.
- Managed Linux server environments for Java application hosting, deployment, and monitoring.
- **Tech Stack:** Java, Spring Boot, Spring WebFlux, JPA, REST APIs, Kafka, AWS Lambda, SNS, SQS, Linux

Projects

IRIS - Interswitch Proprietary Core Banking Payment Gateway (*Internal project - repository not public*) - Java 21, Spring Boot 3.x, Netty TCP, gRPC-Java, Protobuf, Azure Service Bus, Kubernetes

- Built a Netty-based TCP server for the IRIS payment engine and architected ISO 20022/SWIFT inbound/outbound integration for pacs.008, pacs.002, and pacs.028 with C14N canonicalisation, digital signatures, and production error recovery.
- Engineered gRPC-Java RPC validation with Protobuf service contracts across Interswitch's proprietary core banking ecosystem.

Rivet AI - Film Pre-Production SaaS Platform [[Website](#)] - Java, Spring Boot, MongoDB, REST APIs, OAuth 2.0

- Designed Java/Spring Boot REST APIs with Spring Data MongoDB persistence and OAuth 2.0 / Spring Authorization Server authentication and RBAC for a multi-tenant SaaS platform.
- Integrated a Claude-based AI agent to automate manuscript and script processing across pre-production workflows.

PRM - Interswitch Proprietary Payment Risk Management Engine (*Internal project - repository not public*) - Java 21, Spring AOP/AspectJ, Spring WebFlux, Redis, Docker, Kubernetes

- Architected a non-intrusive Spring AOP/AspectJ risk engine mapping 20+ transaction attributes through custom annotations and CQRS patterns, with graceful degradation and a complete regulatory audit pipeline.

StableX - Crypto Wallet & Blockchain Exchange [[GitHub](#)] [[Live Demo](#)] - Java, Spring Boot, Node.js, gRPC-Java, Solidity, PostgreSQL, Docker, React.js

- Designed a Java/Spring Boot-backed decentralised exchange with gRPC-Java blockchain validation and Solidity smart-contract integration; used Node.js Worker Threads to isolate CPU-intensive cryptographic processing from the main event loop.
- Integrated a Claude-based customer-facing AI assistant into the exchange, supporting user onboarding and customer-service interactions.

Nova Bills - Billing and Payments Application - Java, Spring Boot, JPA, PostgreSQL, SQL Server, React.js

- Architected a full-stack Java/Spring Boot billing platform with JPA persistence, Thymeleaf server-rendered views, and a React.js single-page application.

NeuraFeed - AI-Driven Fintech Platform [[GitHub](#)] [[Live Demo](#)] - AI-enabled financial insights, product prototyping, fintech

- Won 3rd place at the Wema Bank Hackathon in Dec 2025 for a platform delivering personalised financial insights.

Education

Babcock University

Ogun State, Nigeria

Bachelor of Science (Honours) in Software Engineering - GPA: 4.10/5.00

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Computer Organisation, Software Architecture, Combinatorics

Certifications

- Oracle Cloud Infrastructure DevOps Certificate, Oracle (Oct 2025)
- Oracle Cloud Infrastructure Developer Certificate, Oracle (Oct 2025)
- Oracle 12c Administrator / Java SE 8 (OCA/OCP), New Horizons (Jul 2024)
- Domain 4 Review: AWS Certified Solutions Architect - Professional (SAP-C02), AWS
- AWS Fundamentals of Architecting on AWS, AWS
- Microsoft Applied Skills: Deploy Cloud-Native Apps Using Azure Container Apps, Microsoft
- Microsoft AI Agentic Cloud Developer Certificate, Microsoft
- Microsoft AZ-204: Developing Solutions for Microsoft Azure (In Progress)
- UiPath RPA Certification, UiPath (Jun 2023)
- Udemy Java Masterclass, Udemy (Jan 2023)

Awards & Affiliations

- 3rd Place - Wema Bank Hackathon (Dec 2025): NeuraFeed, an AI-driven fintech platform delivering personalised financial insights.
- Student Lead, Enactus (Aug 2023) - Digital Onboarders, MindTheGap (Jan 2024) - IEEE Graduate Member